

GRADE 12 BIOLOGY

EUEE MASTER STUDY GUIDE

Ethiopian University Entrance Exam

Complete Exam Preparation

Application · Microorganisms · Energy Transformation · Evolution · Human
Body · Climate Change

Simplified · Exam-Focused · High-Score Strategies

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UNIT 1

APPLICATION OF BIOLOGY

Biology has immense practical applications that shape our lives every day — from conserving nature to producing medicines. This unit links classroom biology to real-world problem-solving.

1.1 Conservation of Natural Resources

Definition — Natural Resource

Any material or energy source from nature that humans use. Resources are classified as renewable (can regenerate) or non-renewable (form over millions of years).

Renewable Resources:

- Solar energy, wind, water (hydropower), forests, soil, wildlife
- These can be replenished if used sustainably

Non-Renewable Resources:

- Fossil fuels (coal, oil, natural gas), minerals, metals
- Once consumed they cannot be replaced within a human lifespan

Biodiversity and its importance:

- **Genetic diversity** — variety within a species (different breeds of cattle)
- **Species diversity** — variety of different species in an area
- **Ecosystem diversity** — variety of habitats and ecosystems

Conservation strategies:

- **In-situ conservation:** protecting species in their natural habitat (national parks, game reserves, biosphere reserves)
- **Ex-situ conservation:** protecting species outside natural habitat (zoos, botanical gardens, seed banks, tissue culture)

EXAM FOCUS

- Know the difference between in-situ and ex-situ conservation with examples.
- Ethiopia's examples: Bale Mountains NP, Simien Mountains NP, Awash NP

1.2 Food and Nutrition Security

Definition — Food Security

The state in which all people at all times have physical, social and economic access to sufficient, safe and nutritious food that meets their dietary needs for an active and healthy life.

Four pillars of food security:

- **Availability** — enough food is produced
- **Access** — people can obtain food (economic/physical access)
- **Utilization** — food is nutritionally adequate and safely prepared
- **Stability** — consistent access over time, not just occasionally

Causes of food insecurity:

- Poverty, drought, poor infrastructure, conflict, population growth, climate change

Biotechnology contributions to food security:

- Development of high-yielding, drought-resistant crop varieties
- Improved livestock breeds with higher productivity
- Biological pest control methods

1.3 Sustainable Development

Definition — Sustainable Development

Development that meets the needs of the present generation without compromising the ability of future generations to meet their own needs. (Brundtland Commission, 1987)

Three pillars:

- **Economic** — growth that creates wealth without environmental destruction
- **Social** — equitable distribution of resources and opportunities
- **Environmental** — protecting ecosystems and natural resources

UN Sustainable Development Goals (SDGs): 17 goals adopted in 2015

- Goal 1: No Poverty | Goal 2: Zero Hunger | Goal 3: Good Health | Goal 13: Climate Action | Goal 15: Life on Land

1.4 Applications in Biotechnology

Definition — Biotechnology

The use of living organisms or their products to modify or improve plants, animals, and microorganisms for specific purposes.

Traditional Biotechnology:

- Fermentation: making bread, beer, wine, cheese using yeast and bacteria
- Selective breeding: improving crop yields and livestock quality

Modern Biotechnology:

- **Genetic engineering:** directly manipulating an organism's DNA
- **Recombinant DNA technology:** combining DNA from different organisms
- **GMOs (Genetically Modified Organisms):** organisms with modified genes for specific traits
- **Cloning:** creating identical copies of organisms or DNA
- **Tissue culture:** growing plants from cells in lab conditions

Medical applications:

- Production of insulin, human growth hormone, vaccines using bacteria
- Gene therapy: correcting defective genes to treat genetic diseases
- Monoclonal antibodies: for cancer diagnosis and treatment

EXAM FOCUS

- Biotechnology applications — both traditional and modern — are heavily tested.
- Know specific examples: insulin from E.coli, HB vaccine from yeast, Bt crop (pest resistance)

PRACTICE QUESTIONS

1. What is the difference between in-situ and ex-situ conservation? Give one example each.
2. List the four pillars of food security.
3. Define biotechnology and give two examples of traditional biotechnology.
4. What are GMOs? Give one advantage and one disadvantage.
5. Explain how recombinant DNA technology is used to produce insulin.

UNIT 2

MICROORGANISMS

Microorganisms are living things too small to see with the naked eye. They include bacteria, archaea, fungi, protozoa, and viruses. Understanding them is crucial for medicine, agriculture, and industry.

2.1 Eubacteria (True Bacteria)

Definition — Bacteria

Single-celled prokaryotic microorganisms. They are the most abundant organisms on Earth and play vital roles in nutrient cycling, decomposition, and human health.

Cell structure (what bacteria have):

- Cell wall (peptidoglycan), cell membrane, cytoplasm, ribosomes, nucleoid (circular DNA), plasmids
- Some have: flagella (movement), pili (attachment), capsule (protection)

What bacteria LACK (unlike eukaryotes):

- No membrane-bound nucleus, no mitochondria, no chloroplasts, no endoplasmic reticulum

Gram staining:

- **Gram-positive:** thick peptidoglycan wall → retains purple stain. Example: Staphylococcus, Streptococcus
- **Gram-negative:** thin peptidoglycan + outer membrane → pink/red. Example: E. coli, Salmonella

Bacterial shapes:

- **Coccus (cocci):** spherical (e.g. Staphylococcus)
- **Bacillus (bacilli):** rod-shaped (e.g. E. coli)
- **Spirillum:** spiral/corkscrew (e.g. Helicobacter pylori)
- **Vibrio:** comma-shaped (e.g. Vibrio cholerae)

Bacterial nutrition types:

- **Autotrophs:** make own food (photoautotrophs use light, chemoautotrophs use chemicals)
- **Heterotrophs:** consume organic matter (saprophytes, parasites, mutualists)

Bacterial reproduction:

- **Asexual:** binary fission (one cell splits into two identical cells) — very fast

- **Sexual:** conjugation (transfer of genetic material via pilus), transformation (absorb DNA from environment), transduction (DNA transfer via bacteriophage)

2.2 Archaea

Definition — Archaea

Ancient prokaryotes that live in extreme environments. Unlike bacteria, their cell walls lack peptidoglycan. They are neither bacteria nor eukaryotes.

Types of Archaea:

- **Methanogens:** produce methane in swamps and gut of animals (Methanobacterium)
- **Halophiles:** live in extremely salty environments (Dead Sea, salt lakes)
- **Thermophiles:** live in very hot environments (hot springs, deep sea vents)
- **Acidophiles:** live in extremely acidic conditions

Beneficial uses of Archaea:

- Methanogens used in biogas production
- Enzymes from thermophiles used in PCR (Taq polymerase) and industrial processes

2.3 Fungi

Definition — Fungi

Eukaryotic organisms that absorb nutrients from their environment. They are neither plants nor animals. Most are multicellular; yeasts are unicellular.

General characteristics:

- Cell wall made of chitin (not cellulose like plants)
- Body made of thread-like hyphae; mass of hyphae = mycelium
- Heterotrophs: saprophytes (decomposers), parasites, or mutualists
- Reproduce sexually and asexually (mainly by spores)

Classification of Fungi:

- **Zygomycetes:** black bread mould (Rhizopus); sexual reproduction by zygospores
- **Ascomycetes (sac fungi):** Penicillium, Aspergillus, Saccharomyces (yeast); reproduce by ascospores
- **Basidiomycetes (club fungi):** mushrooms, toadstools; reproduce by basidiospores
- **Deuteromycetes (imperfect fungi):** no sexual stage known; Candida

Economic importance:

- **Beneficial:** yeast for bread/alcohol, Penicillium for penicillin antibiotic, mushrooms for food, decomposition

- **Harmful:** plant diseases (rust, smut), athlete's foot, ringworm, food spoilage, mycotoxins

2.4 Protozoa

Definition — Protozoa

Single-celled eukaryotic microorganisms. Most are heterotrophs. They are major parasites causing serious human diseases.

Classification by locomotion:

- **Sarcodina (Amoeba):** move by pseudopodia (false feet). Amoeba dysentery (*Entamoeba histolytica*)
- **Flagellates:** move by flagella. Sleeping sickness (*Trypanosoma*), *Giardia*
- **Ciliates:** move by cilia. *Paramecium* (mostly harmless)
- **Sporozoans:** non-motile adults; all are parasites. Malaria (*Plasmodium*)

Key protozoan diseases:

Malaria	<i>Plasmodium</i> sp.	Female <i>Anopheles</i> mosquito
Sleeping sickness	<i>Trypanosoma brucei</i>	Tsetse fly
Amoebic dysentery	<i>Entamoeba histolytica</i>	Contaminated food/water
Giardiasis	<i>Giardia lamblia</i>	Contaminated water
Leishmaniasis	<i>Leishmania</i> sp.	Sandfly bite

2.5 Viruses

Definition — Virus

Non-living infectious particles consisting of genetic material (DNA or RNA) surrounded by a protein coat (capsid). They can only reproduce inside host cells.

Why viruses are considered non-living:

- No cells, no metabolism, cannot reproduce on their own
- Do not respond to stimuli, have no homeostasis

Viral structure:

- **Capsid:** protein coat protecting the genetic material
- **Nucleic acid:** DNA or RNA (never both)
- **Envelope:** some viruses have a lipid envelope (e.g. HIV, Influenza)

Viral symmetry:

- **Icosahedral:** 20-sided, roughly spherical (Adenovirus, Poliovirus)
- **Helical:** rod-shaped helix (Tobacco Mosaic Virus, Rabies)

- **Complex:** bacteriophages (infect bacteria)

Viral replication cycles:

- **Lytic cycle:** virus injects DNA → replicates → lyses (bursts) the host cell → releases virions
- **Lysogenic cycle:** virus DNA integrates into host chromosome (prophage) → dormant → later enters lytic cycle

Important viral diseases:

- HIV/AIDS, Influenza, COVID-19, Hepatitis B, Ebola, Polio, Measles, Rabies, Chickenpox

EXAM FOCUS

- Virus structure, lytic vs lysogenic cycles, and disease examples are all high-frequency EUEE topics.
- Know: HIV is RNA retrovirus, uses reverse transcriptase, attacks CD4+ T-cells.

2.6-2.9 Disease Transmission, Uses and Control of Microorganisms

Modes of disease transmission:

- **Direct contact:** touching, kissing, sexual contact (HIV, herpes)
- **Indirect contact:** contaminated objects (fomites), food, water
- **Airborne:** droplets in air (TB, COVID-19, measles)
- **Vector-borne:** through insects (malaria via mosquito, sleeping sickness via tsetse fly)
- **Vertical:** mother to child during pregnancy/birth/breastfeeding

Beneficial uses of microorganisms:

- Food production: yoghurt, cheese, vinegar, bread, beer, wine
- Antibiotic production: Penicillium → penicillin, Streptomyces → streptomycin
- Bioremediation: cleaning oil spills, breaking down pollutants
- Nitrogen fixation: Rhizobium (in legume root nodules) fixes atmospheric N₂
- Biogas production: methanogens produce methane from organic waste
- Vaccine production, enzyme production, mining (bioleaching)

Controlling microorganisms:

- **Physical methods:** sterilization (autoclave 121°C), pasteurization (72°C for 15 sec), UV radiation, filtration
- **Chemical methods:** antiseptics (skin), disinfectants (surfaces), antibiotics (internal infections)
- **Biological control:** using natural predators or pathogens to control harmful microbes

PRACTICE QUESTIONS

1. What are the differences between bacteria and viruses? List three differences.
2. Explain the lytic and lysogenic cycles of viral replication.
3. Name the pathogen and vector for each: (a) malaria (b) sleeping sickness (c) leishmaniasis.
4. Describe three beneficial uses of microorganisms in industry.
5. What is the difference between antiseptics and disinfectants?

UNIT 3

ENERGY TRANSFORMATION

All living cells need energy to survive. This unit covers how cells capture energy from sunlight (photosynthesis) and release energy from food (cellular respiration).

3.1 Cellular Metabolism

Definition — Metabolism

The sum of all chemical reactions that occur within a living cell. It has two parts: catabolism (breaking down molecules, releasing energy) and anabolism (building up molecules, requiring energy).

Definition — ATP (Adenosine Triphosphate)

The universal energy currency of cells. Energy is stored in the high-energy phosphate bonds. $\text{ATP} \rightarrow \text{ADP} + \text{P}_i$ releases energy for cellular work.

Enzymes in metabolism:

- Biological catalysts that speed up chemical reactions without being consumed
- Have an active site specific to their substrate (lock-and-key model)
- Affected by temperature, pH, substrate concentration, and inhibitors

3.2 Photosynthesis

Definition — Photosynthesis

The process by which green plants, algae, and some bacteria convert light energy, water, and CO_2 into glucose and oxygen. $6\text{CO}_2 + 6\text{H}_2\text{O} + \text{light energy} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$

Where does it happen? Inside chloroplasts.

- **Thylakoid membranes:** light-dependent reactions
- **Stroma:** light-independent reactions (Calvin cycle)

Stage 1: Light-Dependent Reactions (in thylakoid membranes)

- Light energy is absorbed by chlorophyll pigments
- Water molecules are split (photolysis): $2\text{H}_2\text{O} \rightarrow 4\text{H}^+ + 4\text{e}^- + \text{O}_2$
- O_2 is released as a by-product
- ATP and NADPH are produced
- Electron transport chain involves Photosystem I (PSI) and Photosystem II (PSII)

Stage 2: Light-Independent Reactions / Calvin Cycle (in stroma)

- CO₂ is fixed (incorporated) by the enzyme RuBisCO
- ATP and NADPH from light reactions are used
- Glucose (C₆H₁₂O₆) is the end product
- For every 3 CO₂ fixed, 1 molecule of G3P (which becomes glucose) is produced

Photosynthetic pigments (in thylakoids):

- **Chlorophyll a:** primary pigment, absorbs red and blue-violet light, reflects green
- **Chlorophyll b:** accessory pigment, absorbs blue and orange light
- **Carotenoids (carotene, xanthophyll):** accessory pigments, absorb blue-green and blue light

Factors affecting photosynthesis rate:

- Light intensity, CO₂ concentration, temperature, water availability, wavelength of light

3.4 Cellular Respiration

Definition — Cellular Respiration

The process by which cells break down glucose to release energy in the form of ATP. $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + 36-38 \text{ ATP}$

Three stages:

- **1. Glycolysis (cytoplasm):** Glucose (6C) → 2 Pyruvate (3C). Net gain: 2 ATP, 2 NADH. Does NOT require O₂.
- **2. Krebs Cycle / Citric Acid Cycle (matrix of mitochondria):** Pyruvate → Acetyl-CoA → CO₂ released. Produces: 2 ATP, 6 NADH, 2 FADH₂ per glucose.
- **3. Electron Transport Chain/Oxidative Phosphorylation (inner mitochondrial membrane):** NADH and FADH₂ donate electrons. O₂ is the final electron acceptor → H₂O. Produces ~32-34 ATP.

Anaerobic respiration (fermentation) — when O₂ is absent:

- **Lactic acid fermentation:** Pyruvate → Lactic acid. Occurs in muscle cells during intense exercise. Causes muscle fatigue.
- **Alcoholic fermentation:** Pyruvate → Ethanol + CO₂. Occurs in yeast. Used in baking (CO₂ rises bread) and brewing.

Comparison:

O ₂ required?	Yes	No
ATP yield	36-38 ATP	2 ATP only
End products	CO ₂ + H ₂ O	Lactic acid OR Ethanol+CO ₂
Location	Cytoplasm + Mitochondria	Cytoplasm only

EXAM FOCUS

- The difference between photosynthesis and respiration is a classic EUEE question.
- Know: Glycolysis in cytoplasm, Krebs cycle in mitochondrial matrix, ETC on inner membrane.
- Photosynthesis equation: $6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$. Respiration: reverse of this.

QUICK REVISION

- **Photosynthesis:** Light reactions (thylakoid) → ATP, NADPH, O_2 . Calvin cycle (stroma) → glucose
- **Respiration:** Glycolysis (cytoplasm) → Krebs (matrix) → ETC (inner membrane) = 36-38 ATP
- **Anaerobic:** Lactic acid (muscles) or Alcohol+ CO_2 (yeast). Only 2 ATP produced.

PRACTICE QUESTIONS

1. Write the overall equation for photosynthesis. Where do the light-dependent reactions occur?
2. What is the role of ATP in cellular metabolism?
3. Compare aerobic and anaerobic respiration in terms of: (a) oxygen requirement (b) ATP yield (c) products.
4. What happens during glycolysis? Where does it occur?
5. Name two products of the light-dependent reactions that are used in the Calvin cycle.

UNIT 4

EVOLUTION

Evolution explains the diversity of life on Earth. It is the change in inherited characteristics of populations over successive generations.

4.1 Definition and Theories of Evolution

Definition — Evolution

The gradual change in the heritable characteristics of biological populations over successive generations. It results in the formation of new species over long periods of time.

Jean-Baptiste Lamarck (1744-1829) — Theory of Acquired Characteristics:

- Organisms change during their lifetime in response to their environment
- Acquired characteristics are passed to offspring
- Example: giraffes stretched their necks and passed on longer necks
- **Criticism:** acquired characteristics are NOT inherited (Weismann's experiments)

Charles Darwin (1809-1882) — Theory of Natural Selection:

- Organisms produce more offspring than can survive
- There is variation within populations
- Individuals with favourable variations survive and reproduce (survival of the fittest)
- Favourable traits are inherited and become more common over time
- Published in "On the Origin of Species" (1859)

4.1.3 Evidence for Evolution

- **Fossil Record:** shows progressive changes in organisms over geological time. Transitional fossils link ancestral to modern forms.
- **Comparative Anatomy:**
 - Homologous structures: same basic structure, different function (human arm, bat wing, whale flipper — all have same bones). Suggests common ancestry.
 - Analogous structures: different structure, same function (bird wing vs insect wing). Result of convergent evolution.
 - Vestigial structures: remnants of once-functional structures (human appendix, coccyx, wisdom teeth).
- **Comparative Embryology:** embryos of different vertebrates look very similar at early stages, suggesting common ancestry.

- **Molecular Evidence:** similarities in DNA sequences, proteins (cytochrome c) across species indicate evolutionary relationships.
- **Biogeography:** geographic distribution of species (Darwin's finches in Galapagos Islands).

4.1.4 Natural Selection — Types and Examples

- **Directional selection:** one extreme phenotype is favoured. Population shifts in one direction. Example: antibiotic resistance in bacteria.
- **Stabilizing selection:** intermediate phenotype is favoured, extremes are eliminated. Example: human birth weight.
- **Disruptive selection:** both extreme phenotypes are favoured, intermediate eliminated. Example: bill size in finches.

Key concepts:

- **Adaptation:** heritable trait that increases fitness in an environment
- **Fitness:** ability to survive and reproduce in a given environment
- **Speciation:** formation of new species through reproductive isolation

4.1.5 Human Evolution

Human evolution traces the development of the genus *Homo* from primate ancestors. Ethiopia is one of the most important sites for human fossil discoveries.

Key milestones (in order):

- **Australopithecus afarensis:** 3-4 MYA, bipedal, small brain. "Lucy" found in Afar, Ethiopia (1974) by Donald Johanson.
- **Homo habilis:** 2.5 MYA, first tool user ("Handy Man"), larger brain (~650 cc)
- **Homo erectus:** 1.8 MYA, stood fully upright, used fire, migrated out of Africa
- **Homo sapiens:** 0.3 MYA to present, large brain (~1350 cc), complex language and culture

Ethiopian significance:

- "Lucy" (*Australopithecus afarensis*) found in Hadar, Afar region, 1974
- "Selam" (child fossil) found in Dikika, Afar, 2000
- Ethiopia is sometimes called the "Cradle of Humankind"

4.1.6 Mutation and Genetic Drift

Definition — Mutation

A change in the DNA sequence of an organism. Mutations are the ultimate source of new genetic variation.

- **Gene mutation:** change in a single gene (point mutation, frameshift mutation)

- **Chromosomal mutation:** change in chromosome number or structure (deletion, inversion, translocation, duplication)
- **Causes of mutation:** radiation (UV, X-rays), chemical mutagens (carcinogens), errors in DNA replication

Definition — Genetic Drift

Random change in allele frequencies in a small population by chance alone — not due to natural selection. More significant in small populations.

- **Bottleneck effect:** drastic reduction in population size (e.g. disaster) reduces genetic diversity
- **Founder effect:** small group colonises new area, limited genetic variation
- **Gene flow:** movement of alleles between populations through migration

EXAM FOCUS

- Know the order: Australopithecus → Homo habilis → Homo erectus → Homo sapiens.
- "Lucy" found in Ethiopia — very likely tested in EUEE.
- Know 3 types of natural selection with examples.

PRACTICE QUESTIONS

1. Compare Lamarck's theory with Darwin's theory of evolution.
2. Give four types of evidence for evolution.
3. Distinguish between homologous and analogous structures. Give one example of each.
4. Name the types of natural selection and give one example of each.
5. What is the significance of Ethiopia in human evolutionary history?

UNIT 5

HUMAN BODY SYSTEM

The human body has several interconnected systems. This unit covers the nervous system, sense organs, the endocrine system, and homeostasis. These topics are very frequently tested in the EUEE.

5.1 The Nervous System

Definition — Nervous System

The body's communication network that detects stimuli, processes information, and coordinates responses. It has two main parts: the Central Nervous System (CNS) and Peripheral Nervous System (PNS).

The Neuron (nerve cell) — basic unit of the nervous system:

- **Cell body (soma):** contains nucleus and organelles
- **Dendrites:** receive signals from other neurons
- **Axon:** carries impulses away from cell body
- **Myelin sheath:** fatty insulation around axon, speeds up impulse conduction
- **Synapse:** junction between two neurons; signals cross using neurotransmitters

Types of neurons:

- **Sensory neurons:** carry signals FROM receptors TO CNS
- **Motor neurons:** carry signals FROM CNS TO effectors (muscles/glands)
- **Interneurons (relay neurons):** connect sensory and motor neurons within CNS

Action Potential (Nerve Impulse):

- At rest: inside of axon is negative (-70 mV) — resting potential
- Stimulus causes Na^+ to rush in → depolarisation (inside becomes positive)
- K^+ rushes out → repolarisation (returns to negative)
- Signal travels as a wave of depolarisation along the axon

CNS structure:

- **Brain:** cerebrum (thinking/memory), cerebellum (balance/coordination), medulla oblongata (breathing/heart rate)
- **Spinal cord:** pathway between brain and body; controls reflex actions

Reflex Action:

- An automatic, rapid, involuntary response to a stimulus
- **Reflex arc:** Receptor → Sensory neuron → Relay neuron (spinal cord) → Motor neuron → Effector

- Does NOT involve the brain — this is why it is so fast
- Example: withdrawal from a hot object, knee-jerk reflex

5.2 Sense Organs

The Eye — structure and function:

- **Cornea:** transparent outer layer, refracts (bends) light
- **Iris:** coloured ring, controls pupil size to regulate light entering
- **Lens:** focuses light onto retina (can change shape = accommodation)
- **Retina:** contains photoreceptors: rods (dim light/black-white) and cones (bright light/colour)
- **Fovea (yellow spot):** area of greatest visual acuity, highest cone density
- **Blind spot:** optic disc; no photoreceptors; where optic nerve exits
- **Optic nerve:** carries impulses from retina to brain

Eye defects:

- **Myopia (short-sightedness):** eyeball too long; image forms in FRONT of retina. Corrected by concave (diverging) lens.
- **Hyperopia (long-sightedness):** eyeball too short; image forms BEHIND retina. Corrected by convex (converging) lens.
- **Astigmatism:** irregular cornea shape; corrected by cylindrical lens

The Ear:

- **Outer ear:** pinna (collects sound) → external auditory canal → eardrum (tympanic membrane)
- **Middle ear:** ossicles (malleus, incus, stapes) amplify vibrations
- **Inner ear:** cochlea (hearing — fluid-filled, contains hair cells) and semicircular canals (balance)

5.2.6 The Endocrine System

Definition — Hormone

A chemical messenger produced by endocrine glands, secreted directly into the bloodstream, and carried to target organs where it exerts its effect.

Pituitary	GH, FSH, LH, TSH, ADH, Oxytocin	Master gland; controls other glands
Thyroid	Thyroxine	Controls metabolic rate
Parathyroid	PTH (Parathormone)	Regulates blood calcium levels
Adrenal cortex	Cortisol, Aldosterone	Stress response, sodium balance
Adrenal medulla	Adrenaline (Epinephrine)	Fight-or-flight response

Pancreas (islets)	Insulin, Glucagon	Blood glucose regulation
Gonads (ovary)	Oestrogen, Progesterone	Female sex characteristics, reproduction
Gonads (testes)	Testosterone	Male sex characteristics

Diabetes Mellitus:

- **Type 1:** pancreas does not produce insulin; insulin injections required
- **Type 2:** cells resist insulin action; managed by diet, exercise, medication

5.3 Homeostasis and the Kidney

Definition — Homeostasis

The maintenance of a stable internal environment in the body despite changes in the external environment. Examples: regulation of body temperature, blood glucose, and water content.

The Kidney — structure and function:

- Main organ of excretion and osmoregulation
- **Nephron:** functional unit of the kidney; each kidney has ~1 million nephrons
- **Glomerulus:** capillary knot inside Bowman's capsule; filtration of blood under pressure
- **Filtrate:** water, glucose, urea, salts, amino acids (filtered from blood)
- **Tubular reabsorption:** useful substances (glucose, water, amino acids) are reabsorbed into blood
- **Urine:** remaining filtrate = urea, water, salts, creatinine

ADH (Antidiuretic Hormone) and osmoregulation:

- When blood is too concentrated (dehydrated): ADH released → kidney reabsorbs more water → concentrated urine
- When blood is dilute (overhydrated): less ADH → less water reabsorbed → dilute urine

EXAM FOCUS

- Endocrine glands and their hormones is a guaranteed EUCE topic — memorise the table.
- Kidney function: filtration → reabsorption → secretion → urine formation.
- Eye defects: myopia/concave lens; hyperopia/convex lens. Always match defect to lens type.

PRACTICE QUESTIONS

1. Draw and label a nephron. Explain what happens at each part.
2. What is the role of ADH in controlling urine concentration?
3. Distinguish between the roles of rods and cones in the eye.
4. Name the hormone produced by each: (a) pancreas β -cells (b) adrenal medulla (c) thyroid gland.
5. Explain the difference between the nervous system and endocrine system responses.

UNIT 6

CLIMATE CHANGE

6.1 Causes of Climate Change

Definition — Climate Change

A long-term shift in global or regional climate patterns. Primarily refers to the rapid changes since the mid-20th century driven by human activities that increase greenhouse gases.

Natural causes:

- Volcanic eruptions (release CO₂ and sulphur dioxide)
- Changes in solar output (Milankovitch cycles)
- Natural variation in Earth's orbit and tilt

Human (anthropogenic) causes — main drivers:

- **Burning fossil fuels:** CO₂ and other GHGs released from coal, oil, gas
- **Deforestation:** trees absorb CO₂; cutting them increases atmospheric CO₂
- **Agriculture:** rice paddies produce methane (CH₄); cattle produce methane; fertilizers produce N₂O
- **Industrial processes:** cement, steel, chemical manufacturing
- **Landfills:** decomposing waste produces methane

Greenhouse gases (GHGs): CO₂, CH₄ (methane), N₂O, water vapour, CFCs

- GHGs trap heat in the atmosphere (greenhouse effect)
- CO₂ is the most significant human-produced GHG

6.2 Effects of Climate Change

- **Rising temperatures:** global average has increased ~1.1°C since pre-industrial times
- **Melting glaciers and ice caps:** sea levels rising, threatening coastal areas
- **Extreme weather events:** more frequent droughts, floods, cyclones, heatwaves
- **Biodiversity loss:** species unable to adapt fast enough go extinct; coral bleaching
- **Agricultural impacts:** reduced crop yields in tropical regions, shifting growing seasons
- **Human health:** spread of diseases (malaria expanding), heat-related illness
- **Ocean acidification:** CO₂ dissolves in water → carbonic acid → harms marine life (corals, shellfish)

6.3 International Conventions

- **UNFCCC (1992):** United Nations Framework Convention on Climate Change — international framework for addressing climate change
- **Kyoto Protocol (1997):** set binding emission reduction targets for developed countries; introduced carbon trading
- **Paris Agreement (2015):** countries committed to limit global warming to well below 2°C above pre-industrial levels (target 1.5°C)
- **CBD (Convention on Biological Diversity, 1992):** protect biodiversity globally

Ethiopia's response:

- Climate Resilient Green Economy (CRGE) strategy — aims for middle-income status without increasing emissions
- Reforestation programmes (Green Legacy Initiative)

QUICK REVISION

- **GHGs:** CO₂, CH₄, N₂O, CFCs, water vapour
- **Effects:** warming, sea level rise, extreme weather, biodiversity loss, food insecurity
- **Conventions:** UNFCCC (1992) → Kyoto (1997) → Paris (2015)
- **Ethiopia:** CRGE strategy, Green Legacy Initiative

PRACTICE QUESTIONS

1. Distinguish between weather and climate.
2. List five human activities that contribute to climate change.
3. What is ocean acidification and how does it affect marine organisms?
4. What was agreed in the Paris Agreement of 2015?
5. Describe two effects of climate change on biodiversity.

LAST-MINUTE REVISION GUIDE

Photosynthesis eq.	$6\text{CO}_2 + 6\text{H}_2\text{O} + \text{light} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$
Respiration eq.	$\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \rightarrow 6\text{CO}_2 + 6\text{H}_2\text{O} + 36\text{-}38 \text{ ATP}$
Glycolysis	Cytoplasm; 2 ATP net; glucose $\rightarrow$ pyruvate
Krebs Cycle	Mitochondrial matrix; CO_2 released; NADH, FADH_2 produced
ETC	Inner mitochondrial membrane; O_2 used; 32-34 ATP produced
Bacteria vs Virus	Bacteria: cells, can reproduce alone, killed by antibiotics. Virus: not cells, needs host.
Malaria pathogen	Plasmodium; vector: female Anopheles mosquito
HIV	RNA retrovirus; attacks CD4^+ T cells; transmitted by blood/sex/mother-child
Lucy	Australopithecus afarensis, 3.2 MYA, found in Hadar, Afar, Ethiopia, 1974
Natural selection	Variation $\rightarrow$ Selection pressure $\rightarrow$ Survival of fittest $\rightarrow$ Inheritance $\rightarrow$ Change
Myopia fix	Concave (diverging) lens
Hyperopia fix	Convex (converging) lens
Insulin source	Beta cells of pancreatic islets of Langerhans
Homeostasis eg	Blood glucose, body temperature, water balance
In-situ eg	National parks, game reserves, biosphere reserves
Ex-situ eg	Zoos, botanical gardens, seed banks
Kyoto Protocol	1997; binding emission cuts for developed countries
Paris Agreement	2015; limit warming to $1.5\text{-}2^\circ\text{C}$

EUEE EXAM STRATEGY

Time Management

- **Skim first (5 min):** Read all questions. Mark easy (E), medium (M), hard (H).
- **Easy first:** Answer all E questions to bank safe marks and build confidence.
- **Pace yourself:** Aim for ~2 minutes per question. If stuck over 3 minutes, skip and return.
- **Show working:** Even wrong answers earn partial marks if steps are correct.
- **Final review:** Use last 10 minutes to check calculations, names, and dates.

Answering Techniques

- Multiple-choice: eliminate obviously wrong answers first, then choose from what remains.
- Definition questions: always include the key term in your answer.
- Explanation questions: structure as Definition → Example → Importance.
- Diagram questions: label all parts, use arrows correctly, add units where needed.
- Essay questions: write a clear intro, 3-4 main points with evidence, strong conclusion.

Common Exam Traps

- Confusing similar terms — always define the exact term asked.
- Incomplete answers — re-read every question before finalising.
- Time overrun — if a question takes too long, note it and move on.
- Skipping diagrams — a well-labelled diagram often earns more marks than a paragraph.
- Not reading all options in MCQ — always read all four choices before selecting.

NEXT LEVEL LEARNING

Molecular Biology & Genomics

At university you will study gene expression in detail: DNA replication, transcription (DNA→RNA), and translation (RNA→Protein). The Human Genome Project mapped all 3.2 billion base pairs of human DNA.

Immunology

The immune system has two lines: innate (general) and adaptive (specific). B-cells produce antibodies; T-cells attack infected cells. Monoclonal antibodies are used in cancer treatment (checkpoint inhibitors).

Neuroscience

Advanced study of the brain covers neural networks, neuroplasticity, brain imaging (fMRI), and how memory, learning and consciousness work at a molecular level.

Synthetic Biology

Scientists can now design and build new biological systems — creating entirely new microorganisms for medicine, biofuel, and environmental clean-up.